Lecture 26

Higher homotopy groups and topological defects

Objectives. *A loop detects whether a line defect can be removed while the surrounding medium stays ordered. A sphere provides the corresponding test for a point defect. We extend the fundamental group to maps from higher-dimensional spheres. We then distinguish based charges from the free homotopy classes that describe local defects. This distinction identifies opposite point charges for unoriented axes, but preserves their sign for oriented magnetic directions.*

By the end of this lecture you will be able to:

- Prove the group laws and commutativity of higher homotopy groups.
- Use the cited sphere groups and fibration sequence in the stated cases.
- Choose the linking circle for a line defect and sphere for a point defect.
- Compute vortex, nematic and sphere-valued charges with the required basepoint qualification.

Unit 26A Sphere maps and their groups

26.1 Ordered values and two defects

Definition 26.1 (Order parameter)**.** An *order parameter* records the local ordered state of a medium.

Definition 26.2 (Order-parameter space)**.** The allowed ordered values form the *order-parameter space* $\mathcal{M}$.

When a symmetry group G acts transitively on the uniform ordered states, Lecture 8 identifies this space with G/H, where H fixes a reference state.

Definition 26.3 (Director)**.** A director is an unoriented unit axis: $\mathbf{n}$ and $-\mathbf{n}$ describe the same state.

Definition 26.4 (Texture)**.** The word *texture* means a continuous spatial configuration of the order parameter.

Physical input. We model a superfluid by a complex field whose ordered value is its phase in S^1. We model a Heisenberg ferromagnet by a unit direction in S^2. We model a uniaxial

nematic by a director in $\mathbb{R}\mathrm{P}^2$. These descriptions restrict the field to ordered values away from defect cores.

Definition 26.5 (Defect core)**.** A *defect core* is a region where the chosen ordered description fails.

Outside the cores the order parameter is a continuous map into $\mathcal{M}$.

Definition 26.6 (Linking circle and sphere)**.** A small circle surrounding a line in three-dimensional space is a *linking circle*. A small sphere surrounding an isolated point is its *linking sphere*.

Definition 26.7 (Codimension and transverse disk)**.** The *codimension* of a smooth core is the spatial dimension minus the core dimension. A *transverse disk* meets the core at one point in directions complementary to its tangent directions.

The figure separates these two domains of the order-parameter map.

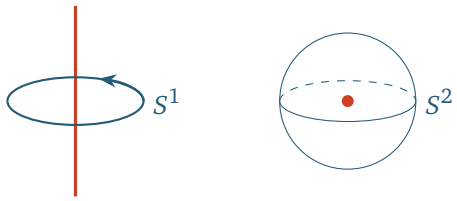


Figure 26.1: A circle links a line core, while a sphere encloses a point core. The order parameter is required to remain in $\mathcal{M}$ on the linking domain throughout a deformation.

For a straight superfluid core, a counter-clockwise linking circle can carry $\phi \mapsto e^{i\phi}$. Its winding is +1 by Lecture 3. It cannot extend to a continuous phase on the transverse disk because such an extension would contract that loop. For a magnetic point core, the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ on $\mathbb{R}^3 \setminus \{0\}$ restricts to the identity direction map on an outward linking sphere. Its degree is +1 by Lecture 14. We need sphere maps to describe this point obstruction.

A model energy explains the restriction to ordered values.

Physical input. For a complex field ψ on a spatial region U, take the model energy

$$E[\psi] = \int_U \big(\kappa|\nabla\psi|^2 + W(|\psi|)\big)\mathrm{d}^d x, \qquad W(R) = c(R^2 - \psi_0^2)^2, \qquad \kappa, c, \psi_0 > 0.$$

Here d is the spatial dimension and $R \geq 0$ is the field amplitude.

The potential vanishes precisely on the circle $|\psi| = \psi_0$. A fixed displacement of the amplitude from that circle raises the potential energy. At fixed nonzero amplitude, spatial variation of an initially uniform phase raises the gradient energy. A spatially uniform phase shift leaves this energy unchanged. A continuous nonzero complex field on a disk has a continuous normalized phase there, so its boundary winding is zero. Nonzero boundary winding therefore forces a zero of the complex field inside. Topology does not determine the size or lifetime of the region where the amplitude departs from ψ_0. The figure shows the potential and an explicit trial core profile.

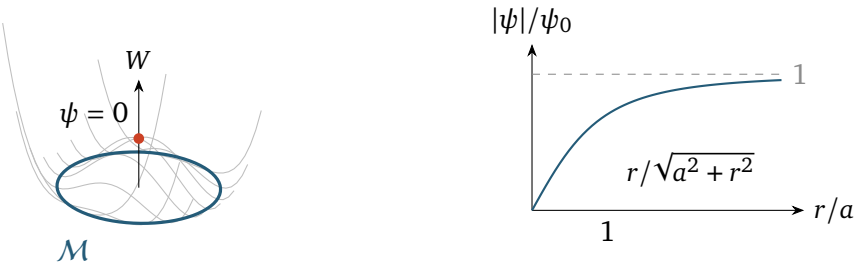


Figure 26.2: The model potential has a circle of minima. The trial field $\psi = \psi_0(x + iy)/\sqrt{a^2 + r^2}$, with $r^2 = x^2 + y^2$ and $a > 0$, has a zero at the centre and approaches the ordered amplitude.

26.2 Maps from cubes and higher homotopy groups

Let X be a space with basepoint $x_0 \in X$. For $n \geq 1$, put $I^n = [0,1]^n$.

Definition 26.8 (Higher homotopy group). The *higher homotopy group* $\pi_n(X, x_0)$ is initially the set of continuous maps $f\colon I^n \to X$ sending the entire boundary to x_0, modulo homotopies that keep that boundary at x_0.

Collapsing the boundary identifies $I^n/\partial I^n$ with S^n. Thus the same classes are based maps from a sphere. The quotient figure marks the entire boundary as one point.

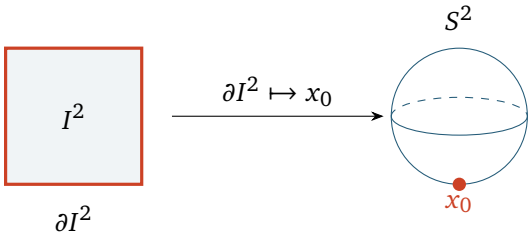


Figure 26.3: Collapsing the square boundary produces a sphere. All boundary points have one image, which is the sphere's basepoint.

For $n = 1$ this is Lecture 3's fundamental group.

Definition 26.9 (Path components). The symbol $\pi_0(X)$ instead denotes the set of path components. It is not generally a group.

Write $t' = (t^2, \ldots, t^n)$. Concatenation in the first coordinate is given by

$$(f+g)(t^1, t') = \begin{cases} f(2t^1, t'), & 0 \leq t^1 \leq 1/2, \\ g(2t^1 - 1, t'), & 1/2 \leq t^1 \leq 1. \end{cases}$$

Both seam values are x_0, so pasting gives a continuous map relative to the boundary. For $n = 1$, this operation traverses f first, in the course's path convention. The plus sign denotes concatenation even when the group is nonabelian. We write γ^{-1} for the reversed path denoted $\bar{\gamma}$ in Lecture 3.

Proposition 26.10 (The group structure · *proved*)**.** *Concatenation makes $\pi_n(X, x_0)$ a group for every $n \geq 1$. Its identity is the constant map and its inverse is represented by $f^-(t^1, t') = f(1 - t^1, t')$. For $n \geq 2$ the group is abelian.*

Proof. Pasting homotopies on the two halves shows that concatenation is well-defined on classes. The two bracketings of three maps occupy first-coordinate slabs of lengths $(1/4, 1/4, 1/2)$ and $(1/2, 1/4, 1/4)$. For a homotopy parameter $u \in [0, 1]$, use slab lengths $((1+u)/4, 1/4, (2-u)/4)$ and rescale the first coordinate on each slab affinely to the unit interval. All lengths are positive. Adjacent maps agree at their boundary value x_0. This continuous reparametrization proves associativity.

A constant first slab is a pause before f. Shrink its length a continuously from $1/2$ to zero, using the value x_0 for $t^1 \leq a$ and $f((t^1 - a)/(1 - a), t')$ for $t^1 \geq a$. This is a homotopy relative to the cube boundary and proves the left unit law. Shrinking a final pause proves the right unit law. For $f + f^-$ the explicit retracing contraction is

$$K_u(t^1, t') = f\big((1 - u)\min\{2t^1, 2 - 2t^1\}, t'\big).$$

At $u = 0$ it is the concatenation and at $u = 1$ it is the constant boundary value. For $f^- + f$, replace the first argument by $1 - (1 - u)\min\{2t^1, 2 - 2t^1\}$, ending at the other boundary face. This proves both inverse laws.

For $n \geq 2$, let $\star$ concatenate in the second coordinate. The same proof gives it the same identity, denoted 0 at the class level. Pasting four maps into quadrants gives the exact interchange identity shown in the figure:

$$(a + b) \star (c + d) = (a \star c) + (b \star d).$$

Here $a, b, c, d \in \pi_n(X, x_0)$. Its two specializations give

$$a{+}b = (a{\star}0){+}(0{\star}b) = (a{+}0){\star}(0{+}b) = a{\star}b, \qquad a{+}b = (0{\star}a){+}(b{\star}0) = (0{+}b){\star}(a{+}0) = b{\star}a.$$

The operations therefore agree and are commutative. ∎

The figure uses only the first two coordinates. The other coordinates stay fixed.

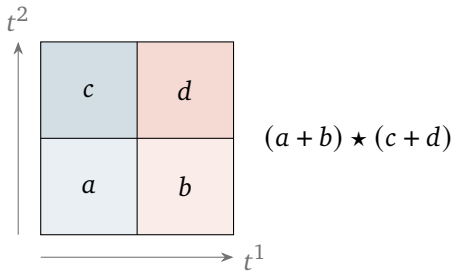


Figure 26.4: The four cube maps have the same placement whether one first joins horizontal pairs or first joins vertical pairs. All seam faces map to the basepoint.

For example, based maps $S^2 \to S^2$ add by joining two sphere maps at their basepoint. The next section identifies this group law with addition of degrees. The extra coordinate also gives a geometric explanation of commutativity. We can compress the two maps into disjoint small squares, with value x_0 outside them. We move the squares around each other in the first two coordinates. All other coordinates stay fixed. There is no second coordinate

for this interchange when $n = 1$. The figure shows disjoint square positions throughout the exchange.

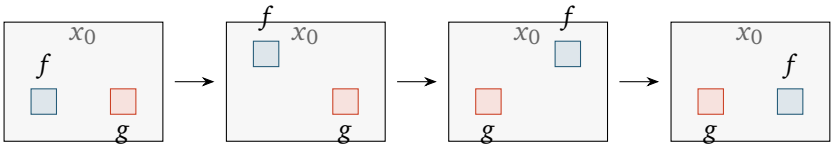


Figure 26.5: The square carrying f moves above the square carrying g before they exchange horizontal positions. The complement maps to x_0 throughout.

26.3 Sphere groups and the Hopf map

The degree from Lecture 14 is invariant under homotopy. The converse for sphere maps is an additional theorem: equal degrees imply based homotopy. We cite it here rather than infer it from the integral formula alone.

Theorem 26.11 (Homotopy groups of spheres · *cited*). *For every $n \geq 1$, the degree map gives $\pi_n(S^n) \cong \mathbb{Z}$. For $1 \leq k < n$, the group $\pi_k(S^n)$ vanishes. The group $\pi_3(S^2)$ is infinite cyclic and is generated by the Hopf map $S^3 \to S^2$ of Lecture 20.*

The precise sources are Hatcher, Corollary 4.9, p. 349, Corollary 4.25, p. 361, and Example 4.51, p. 380. The degree is Lecture 14's integral degree.[1] Thus the identity sphere map has charge +1 and generates $\pi_n(S^n)$. For smooth representatives constant near the cube boundary, choose a common regular value different from x_0. Its preimages in $f + g$ lie in the two half-cubes. Both first-coordinate rescalings preserve orientation. The signed preimage counts therefore give

$$\deg(f + g) = \deg f + \deg g.$$

We first compress the cube map into its interior to make it constant near the boundary. We then use smooth approximation relative to that neighbourhood (Lee, Theorem 6.26, recalled in Lecture 13).

A smooth map $S^k \to S^n$ with $k < n$ misses a point: its entire image consists of critical values and has measure zero by Sard's theorem recalled in Lecture 14. The complement of a point in S^n is contractible by stereographic projection. The contraction can fix the image basepoint, so the map is based null-homotopic. Smoothing supplies the same conclusion for

[1] Hatcher's homological degree is the integer multiplier d on $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$. This definition is given in §2.2, p. 134. This multiplier does not depend on a choice of generator. For smooth maps, Lecture 14's cited de Rham identification intertwines this action with pullback of forms: integrating a form over a composed simplex equals integrating its pullback over that simplex. Thus the homological degree agrees with the integral degree of Theorems 14.3 and 14.14. To verify this, let c be the period of a normalized volume form ν on a generator $[S]$. Its de Rham class is nonzero since $\int_{S^n} \nu = 1$, so injectivity of the pairing gives $c \neq 0$. The top-cohomology argument in Lecture 14 gives $[f^*\nu] = (\deg f)[\nu]$. Naturality and the vanishing of exact-form periods therefore give

$$dc = \int_{f_*[S]} \nu = \int_{[S]} f^*\nu = (\deg f)c.$$

Cancellation of c identifies the two degrees. For continuous maps, Hatcher's definition uses homology directly.

continuous maps. There is no missing-point argument for the Hopf map, which is onto S^2. Its nontrivial class in $\pi_3(S^2)$ is the cited higher-dimensional example.

Lecture 3's winding belongs to $\pi_1(S^1)$. Lecture 14's index of a zero in dimension $n \geq 2$ is the degree of a map in $\pi_{n-1}(S^{n-1})$. Lecture 22's transition winding belongs to $\pi_1(\mathrm{U}(1))$. Lecture 23's integer k is the degree of the asymptotic map $S^3 \to \mathrm{SU}(2)$, a class in $\pi_3(\mathrm{SU}(2)) \cong \mathbb{Z}$ under that lecture's orientation convention.

We will also use the covering-space result in Hatcher, Proposition 4.1, p. 342: a covering induces an isomorphism on π_n for $n \geq 2$. Sphere maps and their homotopies lift, which explains this result. The contractible cover $\mathbb{R} \to S^1$ gives $\pi_n(S^1) = 0$ for $n \geq 2$. The cover $S^2 \to \mathbb{R}\mathrm{P}^2$ gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$ after choosing a lifted basepoint. That choice will matter when the basepoint is allowed to move.

26.4 The fibration sequence and two consequences

Definition 26.12 (Fibration). A *fibration* is a continuous map $p\colon E \to B$ with the homotopy lifting property: for every space Y, a homotopy $Y \times [0,1] \to B$ with a chosen lift at its initial time has a continuous lift throughout that restricts to the chosen lift at time zero.

For the theorem below it suffices to require this property for disks of every dimension. Fibre bundles have this property for disks by Hatcher's Proposition 4.48, p. 379. Choose a basepoint $b_0 \in B$ and a point e_0 in its fibre $F = p^{-1}(b_0)$. All homotopy groups below use these compatible basepoints.

Theorem 26.13 (The fibration sequence · *cited*). *For a fibration over a path-connected base, the homotopy groups have a long exact sequence. Its relevant segment is*

$$\cdots \longrightarrow \pi_2(F) \longrightarrow \pi_2(E) \longrightarrow \pi_2(B) \xrightarrow{\partial_{\mathrm{fib}}} \pi_1(F) \longrightarrow \pi_1(E) \longrightarrow \pi_1(B) \longrightarrow \cdots .$$

Exactness means that the image of each arrow is the kernel of the next.

This is Hatcher's Theorem 4.41, p. 376. The symbol ∂_{fib} names the connecting homomorphism, distinct from a geometric boundary. Represent a based sphere by a square with its boundary collapsed. Lifting this square leaves a boundary loop in the fibre. Its construction and the general exactness proof are cited. The figure distinguishes the collapsed boundary in the base from its lifted loop in the fibre.

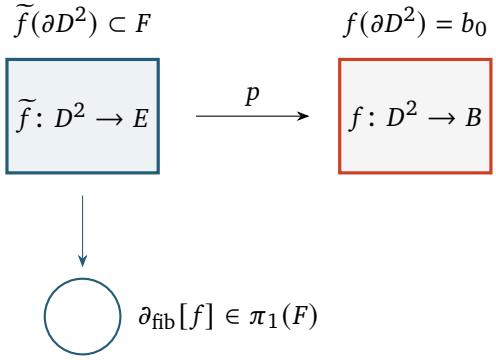


Figure 26.6: A lifted disk has boundary in the fibre because the base disk boundary maps to b_0. That fibre loop represents $\partial_{\mathrm{fib}}[f]$.

For the Hopf bundle, $F = S^1$, $E = S^3$ and $B = S^2$. The cited sphere groups give $\pi_2(S^3) = 0$ and $\pi_1(S^3) = 0$. The figure isolates the resulting segment.

$$0 = \pi_2(S^3) \longrightarrow \pi_2(S^2) \xrightarrow[\cong]{\partial_{\mathrm{fib}}} \pi_1(S^1) \longrightarrow \pi_1(S^3) = 0$$

Figure 26.7: Exactness makes the Hopf connecting map an isomorphism: its kernel is zero and its image is the entire fibre-loop group.

Hence $\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$. The connecting map gives more information than the abstract integer group: it assigns a fibre loop to a sphere map. For the Hopf bundle, a degree-n sphere map gives a fibre loop of winding $\pm n$. The sign depends on the orientation used to define the connecting map. Lecture 22 instead fixes a sign by its explicit northern and southern frame order when it computes the winding of the equatorial transition.

More generally, let G be a simply connected Lie group with $\pi_2(G) = 0$ and let H be a closed Lie subgroup. Lecture 20 gives the principal bundle $H \to G \to G/H$. Exactness gives $\pi_2(G/H) \cong \pi_1(H)$ because both adjacent groups from G vanish. Here the basepoint of H is its identity, so its loops lie in its identity component. Both hypotheses $\pi_2(G) = 0$ and $\pi_1(G) = 0$ are needed for this exact-sequence argument. For $G = \mathrm{SU}(2) \cong S^3$ and $H = \mathrm{U}(1)$, both vanish by Theorem 26.11. The quotient is $\mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$. Thus its point charge corresponds to a loop in the subgroup that fixes a direction. Lecture 27 uses this quotient for the Yang–Mills–Higgs monopole.

Self-check. State in one sentence what $\pi_n(X, x_0)$ is and why we introduced it for defects.

Answer. The group $\pi_n(X, x_0)$ consists of based homotopy classes of based maps $S^n \to X$, so it records whether a linking sphere map can contract while its image basepoint stays fixed. A nonzero class obstructs extension over the transverse disk.

Unit 26B Free defect classes and based charges

26.5 Nematic lifts and moving basepoints

A director map from a linking S^2 lifts to the internal unit-vector sphere because its domain is simply connected, by Lecture 4's lifting theorem. There are exactly two lifts, $\widetilde{f}$ and $-\widetilde{f}$. They are maps into the internal sphere, not tangent vector fields on the physical sphere. The antipodal map on the target has degree -1, so these lifts have degrees d and $-d$. The physical linking sphere keeps its outward orientation. The radial director field therefore has both an outward and an inward vector lift. The spatial comparison loop in the figure carries a half-turn of the director. The figure separates the source orientation from the target sign.

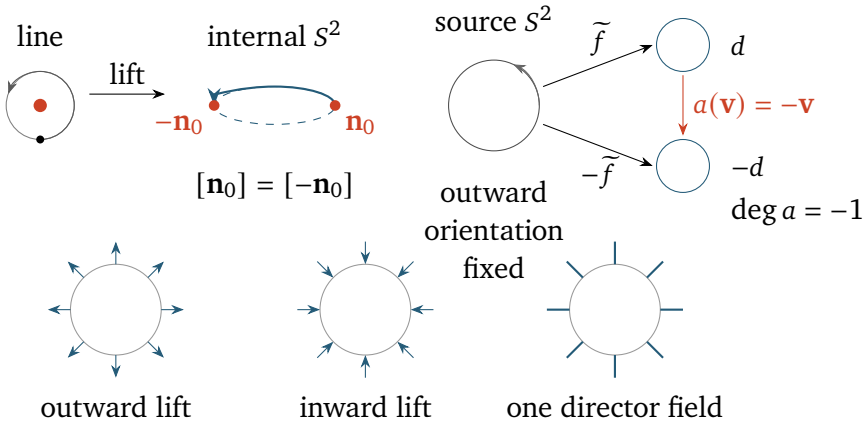


Figure 26.8: A nontrivial director loop lifts from $\mathbf{n}_0$ to $-\mathbf{n}_0$. Changing the target lift reverses its degree while the physical source orientation stays fixed.

For an ordinary based charge, a chosen point of the domain sphere must have a prescribed image x_0. A deformation of the physical texture can move that image. We therefore need the effect of moving the basepoint. Let $\gamma\colon [0,1] \to \mathcal{M}$ run from x_0 to x_1. Rescale the domain of a cube map based at x_1 onto a smaller concentric cube. On each radial segment of the surrounding shell, insert γ, running from x_0 at the outer boundary to x_1 at the inner boundary. The figure depicts this construction.

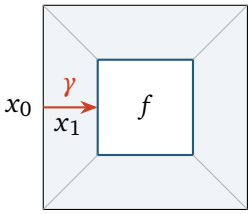


Figure 26.9: The shell changes the inner map's basepoint from x_1 to x_0 along γ. The outer boundary maps constantly to x_0.

Proposition 26.14 (The basepoint action · *sketched*)**.** *The shell construction gives an isomorphism $\beta_\gamma\colon \pi_n(\mathcal{M}, x_1) \to \pi_n(\mathcal{M}, x_0)$. For loops at x_0, it gives an action of $\pi_1(\mathcal{M}, x_0)$ on $\pi_n(\mathcal{M}, x_0)$. For $n = 1$ this is conjugation.*

Sketch. A homotopy of the path or inner map gives a homotopy of the shell map. Reversing the path gives the inverse isomorphism, since the retraced shell contracts. Two nested shells concatenate their paths. Consequently $\beta_{\gamma\eta} = \beta_\gamma \beta_\eta$ in the first-path-first convention. Joining inner cube maps shows that the induced maps preserve the group operation. For $n = 1$ the shell map is the path $\gamma f \gamma^{-1}$, as in Lecture 3. These constructions give the asserted action. They are described in Hatcher, pp. 341–342. ∎

Even though π_n is abelian for $n \geq 2$, this external action can be nontrivial. Commutativity of its own group law does not remove the basepoint issue.

26.6 Local defects and linking spheres

For a core of codimension c, the local transverse boundary is S^{c-1}. This explains which homotopy group occurs. The physical sphere's orientation is fixed. Reversing it can reverse an integer charge.

Definition 26.15 (Topological defect)**.** A *topological defect* has a nontrivial linking homotopy class that obstructs removal of its core within $\mathcal{M}$.

The table distinguishes a linking boundary from the compactification of an entire space.

Space dimension	Core	Codimension	Domain	Datum
3	wall	1	S^0	endpoint components
3	line	2	S^1	loop class
3	point	3	S^2	sphere class
2	line	1	S^0	endpoint components
2	point	2	S^1	loop class
d	none	—	$\mathbb{R}^d \cup \{\infty\} = S^d$	based $\pi_d(\mathcal{M})$

For a wall, the two points of S^0 record an ordered pair of components in $\pi_0(\mathcal{M})$. A nontrivial wall separates distinct components, so this case requires a disconnected target. Lecture 27's kink has this endpoint datum. For line and point defects, free classes require the qualification in the following theorem. The texture row fixes a uniform value at infinity.

Theorem 26.16 (Local defect classes · *sketched*)**.** *Let $\mathcal{M}$ be a path-connected order-parameter manifold. Classify restrictions to a linking domain by continuous homotopy through maps into $\mathcal{M}$. Without a fixed image basepoint, line defects in three dimensions have the conjugacy classes of $\pi_1(\mathcal{M})$, and point defects have the orbits of $\pi_2(\mathcal{M})$ under the $\pi_1(\mathcal{M})$ action. A two-dimensional texture with a fixed uniform value at spatial infinity has a based class in $\pi_2(\mathcal{M})$.*

Sketch. The first assertions concern free homotopy classes of maps from S^1 and S^2. To convert a free homotopy to a based one, follow the moving image of the chosen domain point. Its path is precisely the shell change of basepoint. Thus representatives of two based classes are freely homotopic exactly when this action identifies them. For loops the action is conjugation, recovering Lecture 3's statement: two loops based at x_0 are freely homotopic exactly when their classes are conjugate. For spheres this is its higher-dimensional version (Mermin, §VII.D, p. 631). For the texture, adding the point at infinity compactifies the oriented plane to S^2. The fixed limiting value supplies its target basepoint. This gives the stated based class. ∎

If the value at infinity is also allowed to move, the texture likewise needs action orbits. A nonzero linking class obstructs extension across the core within $\mathcal{M}$: an extension to the transverse disk would contract the boundary map radially. A zero class admits such an extension because a representative map is null-homotopic. This is a topological obstruction, not a claim about the energy or time needed to remove the core. The local boundary classes do not classify knotting or all configurations with several defects.

Moving a linking circle through a region without cores preserves its free class. Indeed, composing the spatial deformation with the order-parameter map gives a homotopy. Fix a common spatial point p and paths η_i from p to two small positive loops ℓ_i. These paths are called *whiskers*. In the arrangement shown, the positive enclosing loop deforms to the ordered loop

$$(\eta_1 \ell_1 \eta_1^{-1})(\eta_2 \ell_2 \eta_2^{-1}).$$

We traverse the right loop first and the left loop second. Composing with the field gives the corresponding ordered product in $\pi_1(\mathcal{M}, x_0)$, where x_0 is the field value at p. For S^1 the integers add. For a nonabelian group, the free conjugacy labels alone do not specify this product. The figure fixes the whiskers and the traversal order.

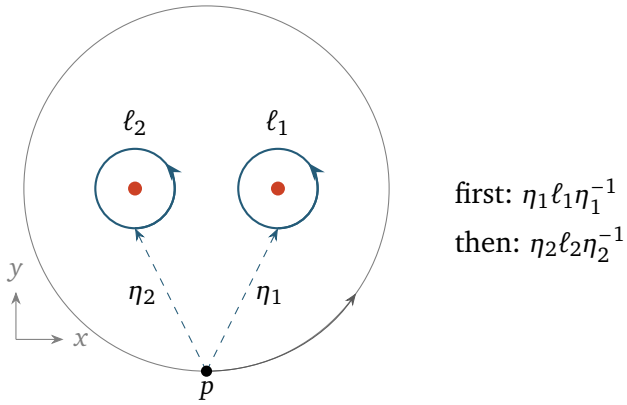


Figure 26.10: All circles run counter-clockwise in the oriented transverse plane. The whiskers give based loops, with loop 1 on the right traversed before loop 2 on the left.

For a concrete nonabelian comparison, take loops in Lecture 5's wedge of two circles. Its fundamental group is the free group F_2 on a and b. The elements b and aba^{-1} are distinct reduced words, but are conjugate by a. They therefore represent one free loop class. This example illustrates the conjugation mechanism independently of a particular medium.

26.7 Vortices, nematics and magnetic textures

The figure compares an oriented phase field with an unoriented director field on a linking circle.

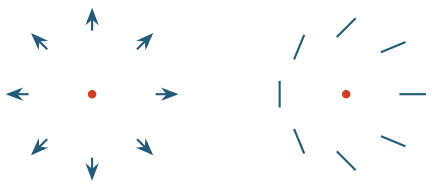


Figure 26.11: The superfluid phase makes a full turn, while the nematic axis closes after a half-turn. Arrows show the phase as a direction in the internal circle, not in space.

Example 26.17 (Superfluid vortices)**.** Determine the local line and point classes for a superfluid. A superfluid has a phase $e^{i\alpha} \in S^1$. On an oriented linking circle a vortex of winding $n \in \mathbb{Z}$ is represented by $\phi \mapsto e^{in\phi}$ for $0 \le \phi \le 2\pi$. Lecture 3 gives $\pi_1(S^1) = \mathbb{Z}$. This group is abelian, so its conjugacy classes are its individual integers. Based joining adds windings. Thus the enclosing winding of two vortices, joined as in the whiskered-loop figure, is the sum of their individual windings. Two vortices of winding +1 have total winding 2 and cannot cancel within S^1. Opposite windings have total zero and have no such obstruction. The group $\pi_2(S^1)$ vanishes, so there is no corresponding point charge.

For a neutral superfluid the phase also determines the circulation.

Physical input. Away from cores, we use the neutral-superfluid relation $\mathbf{v} = (\hbar/m)\nabla\alpha$, where $m > 0$ is the mass of the condensed boson and α is a locally chosen phase.

For $\alpha = n\phi$ on a transverse plane, the velocity is tangential:

$$\mathbf{v} = \frac{n\hbar}{mr}\mathbf{e}_\phi, \qquad \oint \mathbf{v} \cdot \mathrm{d}\boldsymbol{\ell} = \frac{2\pi n\hbar}{m}.$$

Here $r > 0$ is distance from the core and $\mathbf{e}_\phi$ is the counter-clockwise unit tangent. The circulation has the sign of the winding.

Definition 26.18 (Disclination)**.** The nontrivial line defect in a uniaxial nematic is called a *disclination*.

Example 26.19 (Nematic lines and point charges)**.** Determine the local line and point classes for a uniaxial nematic. A uniaxial nematic has $\mathcal{M} = \mathbb{R}\mathrm{P}^2$. Lecture 4 gives $\pi_1(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}_2$. A representative director is $[\cos(\phi/2) : \sin(\phi/2) : 0]$. Its vector lift starts at a unit vector and ends at its negative. The director nevertheless closes, since antipodal vectors give the same axis. Joining two such loops gives a closed full-turn vector loop on S^2. That loop contracts because S^2 is simply connected. Thus the two nontrivial disclinations have total class zero and can cancel topologically.

The chosen sphere cover gives $\pi_2(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$. Changing the lifted basepoint around the nontrivial director loop applies the antipodal deck map. That map has degree -1 on S^2, so its action sends n to $-n$. Hence free point-defect classes identify opposite integers. They can be labelled by $|n|$, including zero. There is no single signed addition law on these orbits without the joining-path data. This is the nematic qualification in Mermin, p. 631. Two local free labels $|1|$ and $|1|$ allow comparison representatives $(+1, +1)$ or $(+1, -1)$ after paths bring them to a common basepoint. Their based sums are 2 and 0. Changing one comparison path by the nontrivial director loop reverses that representative. The local absolute values

therefore do not determine even the magnitude of a total charge. The full configuration and the chosen paths supply additional information. This calculation makes no prediction about the dynamics of bringing the defects together.

A full-turn vector loop can extend across a disk by tilting into the third internal direction. An explicit extension on $x^2 + y^2 \le 1$ is

$$\mathbf{n}_{\mathrm{esc}}(x, y) = \frac{(2x, 2y, 1 - x^2 - y^2)}{1 + x^2 + y^2}.$$

It has unit norm, points along $+\mathbf{e}_z$ at the centre, and equals $(\cos\phi, \sin\phi, 0)$ on the boundary circle. The formula also extends the projected full-turn nematic loop. It does not extend the half-turn disclination generator. The figure shows the internal directions along a radial cross-section.

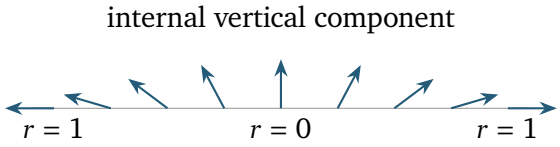


Figure 26.12: The full-turn vector loop has a smooth disk filling in S^2. The arrows tilt to the vertical at the centre, so there is no core.

Definition 26.20 (Hedgehog). The *hedgehog* is the field $\mathbf{n}(\mathbf{r}) = \mathbf{r}/|\mathbf{r}|$ outside the origin.

Definition 26.21 (Skyrmion texture). A *skyrmion texture* is a nonsingular plane texture with a nonzero sphere degree and a fixed uniform value at infinity.

The next figure compares the missing origin of a hedgehog with a smooth texture. For the plane texture, take a scale $a > 0$ and radial distance $r = \sqrt{x^2 + y^2}$. The texture arrows use the formula in the example below.

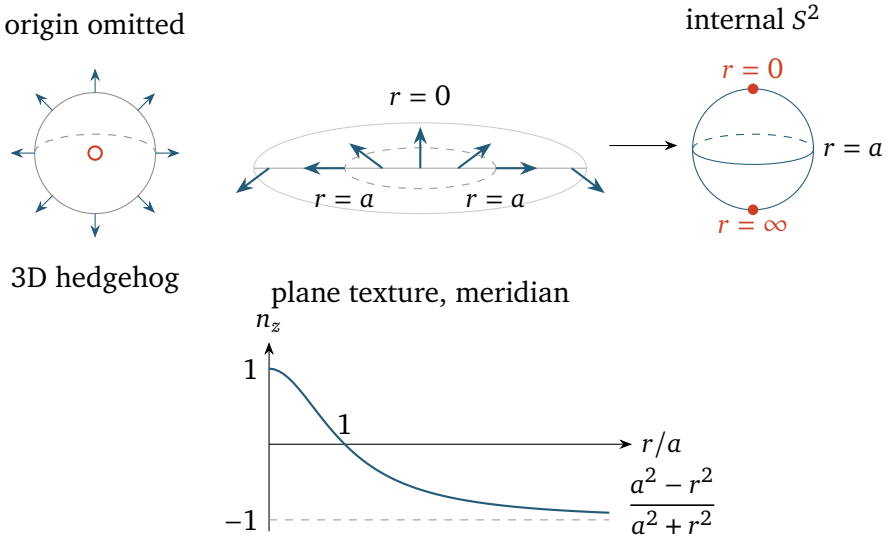


Figure 26.13: The hedgehog has a point core in physical three-space. The plane texture is smooth at its centre and covers the internal direction sphere with degree +1.

Example 26.22 (Magnetic point charge and plane texture)**.** Determine the magnetic point charge and the degree of a compactified plane texture. A Heisenberg ferromagnet has $\mathcal{M} = S^2$. The fundamental group of this target is zero, so there is no local line obstruction of this type. Its point classes are the signed integers $\pi_2(S^2) = \mathbb{Z}$, with no nontrivial basepoint action. The hedgehog restriction to an outward linking sphere is the identity direction map and has degree +1.

For $a > 0$, an explicit degree-one example is

$$\mathbf{n}(x, y) = \frac{(2ax, 2ay, a^2 - x^2 - y^2)}{a^2 + x^2 + y^2}, \qquad \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{4a^2}{(a^2 + x^2 + y^2)^2}.$$

Its norm is one. It tends to the south pole at infinity. With $u = x/a$ and $v = y/a$, this is Lecture 10's inverse stereographic map with its third component reversed. The reflection changes the negative area density in that lecture to the positive density here. The inversion coordinates below are the stereographic chart transition from Lecture 7. For the density calculation, write the field's numerator as $\mathbf{d}$ and denominator as $D = a^2+x^2+y^2$. Differentiating $\mathbf{n} = \mathbf{d}/D$ gives

$$\mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) = \frac{\mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d})}{D^3}, \qquad \mathbf{d} \cdot (\partial_x \mathbf{d} \times \partial_y \mathbf{d}) = 4a^2 D.$$

The terms proportional to $\mathbf{d}$ in either derivative drop out of the determinant. In inversion coordinates $X = x/(x^2 + y^2)$ and $Y = y/(x^2 + y^2)$, the same texture is

$$\mathbf{n} = \frac{(2aX, 2aY, a^2(X^2 + Y^2) - 1)}{1 + a^2(X^2 + Y^2)}.$$

It extends smoothly to $X = Y = 0$, the point at infinity. Its degree is

$$\frac{1}{4\pi} \int_{\mathbb{R}^2} \frac{4a^2 \mathrm{d}x\mathrm{d}y}{(a^2 + x^2 + y^2)^2} = \int_0^\infty \frac{2a^2 r \mathrm{d}r}{(a^2 + r^2)^2} = 1.$$

This is a texture, with no singular core in the plane. The degree labels its topology. No energy-minimizing field equation has been imposed.

Self-check. An ordered medium has $\mathcal{M} = \mathrm{SO}(3)$. Choose the linking domain in three-dimensional space. Find its local line classes and the result of joining two nontrivial lines.

Answer. The linking domain is a circle transverse to the line. Lecture 4 gives $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. It is abelian, so there are two free line classes. The sum of two nontrivial classes is zero, removing this local topological obstruction. This does not assert that a particular physical evolution makes the defects annihilate.

Self-check. A nematic configuration has two point defects. Chosen comparison paths give lifted degrees 3 and −2 at a common basepoint. Find the total based charge and its free label. Repeat when only the second path is changed by the nontrivial director loop. Explain why the local labels 3 and 2 do not determine a total free label.

Answer. The first total is $3 + (-2) = 1$, with free label 1. The changed path acts by the target antipodal map, so the second representative becomes +2. The new total is $3 + 2 = 5$, with free label 5. The same local absolute values have lost the comparison-path information.

Summary. The groups π_n classify based sphere maps and are abelian for $n \geq 2$. The cited sphere results and fibration sequence compute the groups used here. Local free defect classes still require conjugacy for loops and the fundamental-group action for spheres. Superfluid vortices retain signed integer winding. Nematic disclinations have $\mathbb{Z}_2$ charge. Free nematic point charges identify n with $-n$. Sphere-valued hedgehogs and based plane textures have signed degree. Lecture 27 adds boundary conditions and energy to this topological classification.

Reading. Hatcher, §4.1, pp. 340–343 and 349; §4.2, Corollary 4.25, p. 361, Theorem 4.41, p. 376, Proposition 4.48, p. 379, and Example 4.51, p. 380. Mermin, *The topological theory of defects in ordered media*, §VII.D–E, pp. 630–632. Nakahara, §4.8–4.9.